

Project 04: Design a Room

Unit: Unit 4, Spaces We Live In (Environmental Design and Management) **Topic:** 4.3 Design a Room project **Runs across:** Lessons 4.11 (the client brief), 4.12 (the scale plan and traffic flow, two days), 4.13 (the budget), 4.14 (the model, three days), 4.15 (the pitch and gallery walk). Eight days, days 106 to 113. **Team:** alone or a team of two, student's choice with teacher veto **Scored with:** Rubric 04 - Design a Room.md (five criteria, 20 points; the labs and projects category)

Standards this project assesses

- NYS EDM 2. Environment and Interior Design a) Design shared environments and living space zones (for home, school, work, and/or community) which show respect for individual privacy needs and appreciation of large group social needs; b) Create a floor plan designed to meet the needs of individuals, families, or communities; c) Incorporate the elements and principles of design into a design plan; d) Research how environmental and green home factors influence the design of functional, safe and aesthetic living spaces
- NYS EDM 3. Consumer Resources and Finance a) Construct a budget for implementation of a floor plan design; b) Use consumer skills to select household items
- NYS EDM 1. Healthy, Safe, Sustainable Living Spaces b) Identify common safety hazards found in living spaces and describe ways to minimize risks to individuals, families, and household members (wording per the NYSED 2023 copy; the 2018 PDF fused two phrases)
- CTE PSI 2. Design Process (Proactive) a) Implement a formal design process to solve a given problem by a. Defining the problem being addressed; b. Defining criteria that must be met through the finished design; c. Defining constraints that must be adhered to; d. Brainstorming and examining possible solutions; e. Selecting the best solution for evaluation; f. Developing and constructing a prototype or model of the selected design; g. Testing and evaluating the prototype and model against the design criteria and constraints; h. Optimizing the solution for best form and function; i. Evaluating their use of the design process and how it impacted their final solutions
- CTE FCL 4. Buying Goods and Services c) Develop a spending and savings plan (budget) based on income and expenses; f) Apply a decision-making model to maximize consumer satisfaction when purchasing goods and services
- CTE SUS 2. Environmental Management f) Explain factors to consider when assessing the environmental impact of housing decisions
- CTE CIR 1. Communication f) Demonstrate personal development of communication skills through practice of these skills in a variety of classroom applications
- NYS FACS Standard 2 (Intermediate), Key Idea: Students will know the basic principles of home and community safety. They can demonstrate the skills necessary to maintain their homes and workplaces in a safe and comfortable condition. They can provide a

safe and nurturing environment for themselves and others.; PI: Students know the basics of managing a safe and healthy home.

- NYS FACS Standard 3 (Intermediate), PI: Students are able to budget their time and money.
- CDOS 3a Universal Foundation Skills: Thinking Skills; Managing Resources; Technology; Interpersonal Skills
- National FCS 3.0: 11.4.3 Draw an interior space to scale using architectural symbols.



The project sheet (student version)

Name: _____ Date: _____ Team partner (if any): _____ Client:

The job

You are a designer. Somebody has hired you. They have a room, a need, and a budget. Your job is to design a room that solves their problem, prove on paper that it fits and that they can move through it, pay for it without going over, build a model they can hold, and pitch it to them in sixty seconds.

You are not designing your dream room. You are designing theirs.

Two paths

Pick one.

Path A: Design Your Own Bedroom, for a client. One room in a home. The walls, the floor, the door, and the window are already there. Your client needs the room to work for them. Budget: \$400 for furniture, storage, light, paint, and finish. Five of the six client cards are Path A.

Path B: Tiny House. A whole home in 8 feet by 24 feet plus a sleeping loft. A kitchen, a bathroom, a place to work and eat, a place to sit, storage, and the loft bed. Your clients need all of it to fit and to be reachable through a three-foot door. Budget: \$1,500 for the interior. This is the harder path. One client card is Path B.

The six clients

You get one. Read all six before you choose. The full cards, with the room dimensions and where the door, window, and closet are, are on Handout 04.11 - Client Cards and Needs and Wants List.md.

Card	Client	The need in one line	Path	Budget
1	The Okafor family: Grandpa Emeka, 78, uses a walker, is moving into the spare bedroom	A bedroom he can get around with a walker, with nothing to trip on and a bed he can get out of	A	\$400
2	The Delgado family: Mia, 13, and her sister Luz, 8, share a bedroom	Two sleep spots and a study corner for Mia that Luz's toys do not take	A	\$400

		over		
3	The Nguyen family: two parents, two kids in elementary school, a dog, and a small front entry	A mudroom in a 6 by 9 entry: a seat, hooks at kid height, a bin per person, and the door still swings	A	\$400
4	The Ahmed family: Sami and Leyla, a baby due next month, a one-bedroom apartment	Their bed stays; a crib with clear, safe space, a changing spot, and a chair for night feedings	A	\$400
5	Jordan and Sam: a couple who both work on laptops and want a home with no mortgage	A tiny house interior: kitchen, bathroom, a table for two to work and eat, seating, storage, the loft bed, all reachable	B	\$1,500
6	The Rivera family: Marco, 15, uses a manual wheelchair	A bedroom with 36-inch paths, a 5-foot turning circle, a bed and desk at chair height, and a closet he can reach	A	\$400

What you turn in

Deliverable	What it is	When	Rubric criterion
1. The Needs and Wants List and the problem sentence	Every line on the client card sorted into need or want, two hidden needs the card did not say, and one sentence: " <u> </u> needs a room that <u> </u> within <u> </u> ."	Lesson 4.11	1. Meets the client's needs

2. The scale plan	The room on graph paper at 1/2 inch = 1 foot (or 1/4 inch = 1 foot for a tiny house on letter paper), scale written, client named, door swing, windows, closet, every piece of furniture to scale, zones labeled, every path shaded and measured in inches, the turning circle if the client uses a wheelchair, and on the back: two changes made after peer review and one suggestion rejected, each with a reason	Lesson 4.12 Day 2	2. The scale plan and traffic flow
3. The Budget Sheet	Every piece on the plan priced from the catalog with its item number and column (new, secondhand, DIY), a running subtotal, sales tax at 8.625 percent, a total at or under the budget, and three justifications: one new, one secondhand, one DIY or free	Lesson 4.13	3. The budget and consumer choices
4. The model	A shoebox or foam board model (or the isometric drawing alternative) at a scale written on it, walls and floor, door and windows where the plan has them, every piece of	Lesson 4.14 Day 3	4. The model and design principles

	furniture from the plan to scale, the color scheme from Lesson 4.8 applied and named, one texture from the Lesson 4.9 mood board, a GREEN label and a SAFE label each with a one-sentence reason for this client, the client card mounted, and a two-sentence justification card		
5. The pitch	Sixty seconds to your client: the client and the need, the plan and the path, the budget total and the hardest trade-off, the color scheme and the mood, the green and safe features. Every teammate speaks. Then one question from the client, answered in one sentence.	Lesson 4.15	5. The pitch
6. The do-now stack	Every floor plan do now from Lesson 4.10 through Lesson 4.14 Day 3 (eleven sketches), in order, stapled, name on the first sheet	Lesson 4.15	Daily work, not on the rubric; a missing stack is noted on the rubric's notes line

Also in the folder, not scored on the rubric: the four feedback slips you wrote for other teams, and the peer review checklist you filled for another team's plan.

Day by day

Day	Lesson	What happens	Checkpoint
1 (day 106)	4.11	Meet the six clients.	List complete;

		Choose a client, a path, and alone or a partner. Sort needs and wants. Write the problem sentence.	problem sentence names the client, the need, and the limit
2 (day 107)	4.12 Day 1	Draw the room outline to scale from the card. Doors, windows, closet, scale, client name. Zones. Cut out furniture and arrange it. Freeze.	Walls match the card; scale written
3 (day 108)	4.12 Day 2	Shade and measure every path. Check the principles. Peer review with the checklist. Revise. Glue. Write the two changes and one rejection.	Every path passes; teacher initials the plan
4 (day 109)	4.13	Price every piece from the catalog. Running subtotal. Tax. Trade-offs until the total is at or under budget. Three justifications.	Total at or under budget; teacher initials the sheet
5 (day 110)	4.14 Day 1	Mark the plan on the box or base. Cut the door, mark the window, build extra walls. Floor covering.	Box matches the plan
6 (day 111)	4.14 Day 2	Color the walls. Build every piece of furniture to scale from card and scrap. Place it where the plan says.	Furniture matches the plan
7 (day 112)	4.14 Day 3	Finish. GREEN and SAFE labels. Client card mounted.	Clipboard check: matches the plan, scheme named,

		Justification card. Scale written on the model. Self-check criterion 4.	labels give a reason, scale written, justification has a need and a principle
8 (day 113)	4.15	Pitch planner. Pitches in two rounds with a client. Gallery walk, four feedback slips. Self-assessment. Turn everything in.	Folder complete

Where you get to choose

- **The client and the path.** Six cards, two paths. Yours to pick, as long as your table does not double up on a card unless you are a team. You may switch cards until the end of Lesson 4.12 Day 1 with no penalty.
- **Alone or a team of two.** A team turns in one plan, one Budget Sheet, one model, and two do-now stacks. Both teammates speak in the pitch. I can veto a team.
- **The room's answer.** Within the card's numbers and rules, every decision in the room is yours: where the bed goes, what color, new or secondhand, which green choice, which safety feature.
- **The model.** A shoebox, a foam board base, or the isometric drawing. A digital 3-D room in an app the school allows, printed or on a screen at the gallery walk, also counts.
- **The pitch.** Spoken, read from the planner, a picture pitch (point and a teammate reads), or a pitch to me alone at your station. All five parts count the same way.

Supports

- **Sentence starters.** The Spanish lines are to be verified by a native speaker.

English	Spanish
My client is ____.	Mi cliente es ____.
The client needs ____ because ____.	El cliente necesita ____ porque ____.
The path from the door to the bed is ____ inches.	El camino de la puerta a la cama mide ____ pulgadas.
The total is \$____ with tax, under the budget of \$____.	El total es \$____ con impuesto, por debajo del presupuesto de \$____.
I chose ____ secondhand to save money for ____.	Elegí ____ de segunda mano para ahorrar para ____.
The color scheme is ____ and the mood is ____.	El esquema de colores es ____ y el ambiente es ____.
The green choice is . The safe feature is .	La opción verde es . La medida de seguridad es .

- **The picture pitch card.** On the back of the pitch planner: five boxes with an icon each (a person, a floor plan, a dollar sign, a paint chip, a leaf and a shield). A student points to the box, then to the plan or the model, while a teammate or the teacher reads the line. It is scored on the five parts exactly like a spoken pitch.
- **Pre-cut and pre-drawn pieces.** For a student who needs them: the room outline drawn by the teacher, furniture cut-outs pre-cut, wall pieces and door openings pre-cut for the model, bed and desk blanks pre-folded.
- **A shorter list.** Five required pieces of furniture (a bed or crib, one seat, one work or storage surface, one storage piece, one light), one label, a one-sentence justification, three pitch parts. Scored on the same rubric, with the reduction noted on the notes line.
- **Extended time.** Each deliverable may finish in the first five to ten minutes of the next lesson. The whole folder is due at the end of Lesson 4.15; a late project is accepted with a one-level drop on one criterion, never a zero.
- **A larger scale.** 1 inch = 1 foot on 11 by 17 graph paper and a foam board base, for vision or fine-motor needs.
- **Roles in a team of two.** One teammate can own the plan and the model; the other the Budget Sheet, the labels, and the justification. Both speak in the pitch, even if one line.
- **If you are absent** for the model days: the isometric drawing alternative on the build guide is the written alternative, per the grading plan.

How it is scored

Rubric 04 has five criteria, each scored Not Yet (1), Approaching (2), Meets (3), or Exceeds (4), for 20 points. You fill the self-assessment row first with one line of evidence per criterion; then I score. A team's plan, budget, and model earn one score for both teammates. The pitch is scored for the team, with a note if one teammate did not speak.

The project counts in the labs and projects category (40 percent of the marking period). The Unit 4 Test on day 115 has a client scenario, so keep your plan in mind.

Teacher notes on the project

- This is the synthesis project from my Dream Home Challenge packet, rebuilt with a client, a budget, a model, and a pitch, and spread across a 40-minute daily schedule with a checkpoint every day. My packet asked students to design a room of their own; this version asks them to design for someone else, which is the difference between EDM 2. a) (shared space, privacy needs, group needs) and a decoration exercise. The state's own sample learning experiences for FACS include "Design Your Own Bedroom" (6 days) and "Tiny Houses" (9 days); both are offered as paths here in 8 days.
- The six client cards deliberately include a walker user, a wheelchair user, and a new baby, because the state's Health, Safety, and Wellness illustrative activity for this module asks for exactly that: a floor plan revised for a family member who needs special consideration. The numbers on the cards (36-inch path, 60-inch turning circle, bed at chair height, rod at 48 inches) come from the 2010 ADA Standards and common universal design guidance.
- Scale: Lesson 4.10 taught $1/4$ inch = 1 foot. The project plan uses $1/2$ inch = 1 foot so a bedroom fills a sheet of quarter-inch graph paper (two squares per foot). The Unit 4 Test asks both conversions. The model uses the plan's scale or 1 inch = 1 foot, written on the model.
- Budget for the unit plan, per model: shoebox free (collected from students, families, and a shoe store from Lesson 4.6 onward); foam board about \$3 to \$4 per 20 by 30 sheet, two or three bases per sheet if boxes run short; card stock about \$0.30; glue sticks about \$0.25; scrap fabric and paint chips free from a fabric store's remnant bin and a paint department; masking tape and markers from room stock. Target under \$4 per model, realistically \$2 to \$3.50. For 25 models, about \$60 to \$90 if every student needs a foam base, about \$20 if the boxes come in.
- The catalog prices in Handout 04.13 are realistic 2026 figures for Long Island, every one to be refreshed before teaching. The tax math works at any rate; only the teacher key's answers move.
- A team of two is a choice, not a default. In a class that cannot choose well, assign partners by client card and say so on day 1.
- Photograph every model, no names, no faces. They are the website's Unit 4 page and the principal's shelf.